

Calibration: what improved and what needs a decision

Evening follow-up for the September 14 presentation

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The decision in one page

There is a reproducible calibration improvement, but the main housing miss remains. Within the existing parameter bounds, the new candidate lowers loss from 30.483 to **21.798**, a **28.5%** reduction versus retained production and **5.9%** versus the morning candidate. Two exact repeats confirm all twelve fit rows, parameters and the historical path.

Its first-birth housing response is **0.477 rooms against 0.720**: still a 33.8% shortfall. Mean housing remains **6.333 rooms against 5.780**, 9.6% too high. These discrepancies matter more than the improvement in the scalar objective.

Point	Loss	First-birth rooms	Mean rooms
Empirical target	–	0.7202	5.7800
Retained production	30.4830	0.4364	6.3173
Morning candidate	23.1534	0.4660	6.3286
Evening, existing bounds	21.7980	0.4766	6.3325
Observed-gap diagnostic	20.6953	0.5158	6.3491

The last row is a diagnostic fixed-parameter evaluation outside the old first-child bound; the remaining coordinates were not re-estimated. Two exact repeats confirm this point.

The model first-birth response compares rooms in 2023 between matched branches originating with a first birth versus childlessness in 2019. Later births are allowed in the birth branch; the control stays childless. The empirical target remains **0.720246 rooms**. The earlier full-data regression check reproduced it and gave no reason to lower it. Tonight's work keeps all twelve target values and weights, the model equations and the production calibration intact.

My recommendation: keep the empirical target and pursue a joint housing-and-tenure refinement. The final diagnostic improves the first-birth response to **0.516** while matching the family-size room gap, at loss **20.695**. It provides a more promising direction than holding the three-child preference floor fixed. The mean-rooms and ownership gaps remain. Keep policy results tied to the morning calibration until a replacement is chosen and checked.

What the housing experiment changes

The child housing floor is the minimum housing-service component in household preferences. With $m > 0$ children at home, it is

$$\bar{h}(m) = \bar{h}_J + m\bar{h}_C.$$

The first-child term $\bar{h}_J$ adds the same amount for every household with children at home; the slope $\bar{h}_C$ adds more for each additional child. Raising the first term by d and lowering the slope by $d/3$ preserves the three-child floor and raises the one-child floor by $2d/3$. It changes desired housing through an existing model parameter.

The morning candidate has $\bar{h}_J = 0.464931$ and $\bar{h}_C = 0.248877$. Its one-child floor is 0.713808 and its three-child floor is 1.211562. The existing search ceiling $\bar{h}_J = 0.5$ leaves little room for this particular rebalancing. The ceiling applies to a preference parameter; equilibrium housing responses also depend on saving, tenure, fertility choices and market prices.

The 0.5 ceiling entered the August 18 code when the contemporaneous estimate was about 0.20. The bounded history search recovered no empirical or theoretical justification for that endpoint and no direct author choice of it. The experiment README records the historical evidence.

The bounded search completed a 23-case local coordinate panel followed by twelve declared joint cases. It combines the first-child ceiling with alternative per-child floors and supply intercepts. The winning point preserves the three-child floor and combines half-steps along the improving non-housing coordinates.

A lower supply intercept can improve both mean rooms and the first-birth response, but reduces ownership. For example, one joint point gives a 0.502-room response and 6.059 mean rooms, with ownership 0.516 against the 0.575 target; its loss is 23.254. The current winner instead gives 0.477 rooms and ownership 0.547. These are observed trade-offs across the tested points, not a global frontier.

At the bounded winner, the matched birth branch uses **5.807702** rooms and the childless branch **5.331057**, giving the 0.476645 response. Relative to the morning point, the birth branch rises 0.007360 and the control falls 0.003242. Their difference raises the response by only **0.010601 rooms**. This comparison includes changes in equilibrium and the selected risk set; it is not a causal decomposition.

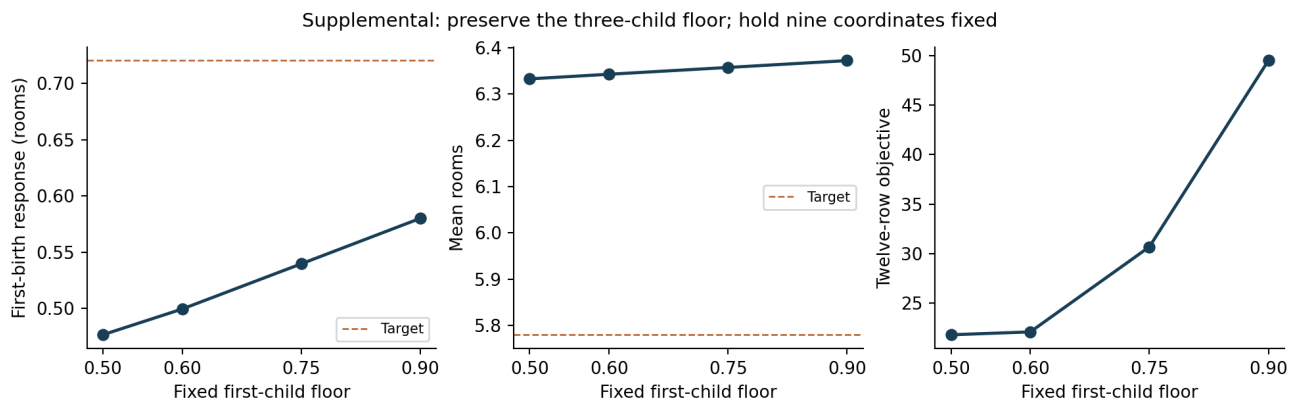


Figure 1: The horizontal axis is the first-child term; the one-child floor equals this term plus the per-child slope. The three-child floor and nine other coordinates are fixed.

Complete fit under the retained search bounds

The best candidate within the existing bounds is the **evening round 2, case 10**, with loss **21.798033**. Its folder is distinct from the retained September 4 task_010. It uses the same twelve-row objective and eleven estimated parameters.

Every gap is model minus target. Loss contributions equal weight times squared gap. Some weights use provisional scales, so the total loss is an objective value, not a chi-squared test statistic.

Moment	Target	Model	Gap	Weight	Loss
Completed fertility, age 42	1.9180	1.9238	0.0058	1425.74	0.0485
Childless share, age 42	0.1880	0.1883	0.0003	17180.7	0.0012
Age at first birth	26.0446	26.3433	0.2987	44.4444	3.9645
First births at age 30+	0.2603	0.2429	-0.0175	10000	3.0453
First-birth rooms response	0.7202	0.4766	-0.2436	137.565	8.1633
Parent rooms: 3+ vs. 1-2	0.3677	0.3727	0.0050	2958.51	0.0742
Parent ownership gap	0.1677	0.1679	0.0002	14229.6	0.0008
Ownership, ages 30-55	0.5755	0.5465	-0.0290	1207.85	1.0130
Mean rooms, ages 18-85	5.7800	6.3325	0.5526	11.9732	3.6557
Wealth / annual earnings	6.8731	7.0374	0.1643	6.28767	0.1698
Bequests / wealth, annual	0.0088	0.0085	-0.0003	5.16529e+06	0.4633
Old wealth/income: p90/p50	3.4481	3.3031	-0.1451	56.9598	1.1984
Total					21.7980

Shares are fractions; ages are years; the housing differences and mean rooms are in rooms. The parent room gap uses ages 30-55. The last row is the ratio of the 90th percentile to the median of wealth/annual income among ages 76-84. Weights are shown to six significant figures; the linked CSVs preserve full precision.

The first-birth housing response accounts for **37.4%** of this loss. Fertility timing contributes another **32.2%**, and mean rooms **16.8%**. Fertility levels and the parent ownership gap fit closely. This concentration suggests focusing the next search on housing demand and birth timing while retaining the full objective. The later-first-birth share fits better than at the morning point, while mean age fits worse; improving one timing statistic alone is insufficient.

First diagnostic: the family-size trade-off

At fixed jumps 0.6, 0.75 and 0.9, first-birth room responses rise to 0.500, 0.540 and 0.580, while losses rise to **22.089, 30.679 and 49.496**. All three points pass the maintained gates and have no occupied value decreases. None improves the full objective. The anchor remains best and was already reproduced twice using the fixed option; two further identical repeats were unnecessary and were not launched.

Preserving the three-child preference floor still allows the observed large-family room gap to fall. At jump 0.9, this gap is **0.275 versus target 0.368**, adding 25.585 loss units relative to the anchor; the parent ownership gap adds another 5.641. The birth-rooms improvement does not offset these costs. This is evidence about the tested path, not proof that the ceiling is irrelevant or that joint re-estimation cannot help.

The complete fit below is for **jump 0.6**, the lowest-loss extended point. The other nine estimated coordinates stay fixed and the per-child slope compensates mechanically; this is not a ten-parameter recalibration.

Moment	Target	Model	Gap	Weight	Loss
Completed fertility, age 42	1.9180	1.9238	0.0058	1425.74	0.0486
Childless share, age 42	0.1880	0.1889	0.0009	17180.7	0.0142
Age at first birth	26.0446	26.3476	0.3029	44.4444	4.0788
First births at age 30+	0.2603	0.2431	-0.0173	10000	2.9787
First-birth rooms response	0.7202	0.4995	-0.2207	137.565	6.7011
Parent rooms: 3+ vs. 1-2	0.3677	0.3477	-0.0200	2958.51	1.1813
Parent ownership gap	0.1677	0.1727	0.0051	14229.6	0.3668
Ownership, ages 30-55	0.5755	0.5457	-0.0298	1207.85	1.0740
Mean rooms, ages 18-85	5.7800	6.3425	0.5625	11.9732	3.7891
Wealth / annual earnings	6.8731	7.0354	0.1623	6.28767	0.1656
Bequests / wealth, annual	0.0088	0.0085	-0.0003	5.16529e+06	0.4697
Old wealth/income: p90/p50	3.4481	3.3017	-0.1464	56.9598	1.2214
Total					22.0892

Follow-up: fit the observed family-size room gap

The last four-point test uses the stable local housing-floor derivatives to choose a per-child slope that approximately preserves the **observed** parent-room contrast. It holds the other nine estimated coordinates fixed, re-normalizes old fertility, then evaluates the complete twelve-row objective. The best point is at a fixed first-child term **0.6** and slope **0.232688**.

It gives a first-birth response of **0.515778 rooms**, and a large-family room gap of **0.367541 against 0.367700**. Loss falls to **20.695274**, 5.1% below the bounded winner. This is a promising diagnostic point outside the original first-child bound; it is not a jointly re-estimated production calibration. **Both exact repeats passed.**

Moment	Target	Model	Gap	Weight	Loss
Completed fertility, age 42	1.9180	1.9238	0.0058	1425.74	0.0483
Childless share, age 42	0.1880	0.1891	0.0011	17180.7	0.0208
Age at first birth	26.0446	26.3492	0.3046	44.4444	4.1230
First births at age 30+	0.2603	0.2431	-0.0172	10000	2.9525
First-birth rooms response	0.7202	0.5158	-0.2045	137.565	5.7512
Parent rooms: 3+ vs. 1-2	0.3677	0.3675	-0.0002	2958.51	0.0001
Parent ownership gap	0.1677	0.1757	0.0081	14229.6	0.9250
Ownership, ages 30-55	0.5755	0.5450	-0.0305	1207.85	1.1240
Mean rooms, ages 18-85	5.7800	6.3491	0.5691	11.9732	3.8784
Wealth / annual earnings	6.8731	7.0338	0.1607	6.28767	0.1624
Bequests / wealth, annual	0.0088	0.0085	-0.0003	5.16529e+06	0.4735
Old wealth/income: p90/p50	3.4481	3.3008	-0.1473	56.9598	1.2361
Total					20.6953

The first-child term is fixed above its original [0, 0.5] interval. The per-child slope is inside [0.1, 1.8] and not near either bound. All other nine coordinates and their restrictions are exactly those in the next page's bounded-parameter table. The normalized old child-value intercept is **0.299775** and its derived 2023 value **-0.027106**. No target or weight changed.

Compared with the bounded point, the birth branch gains 0.0465 rooms and the childless branch gains 0.0073, raising their difference by 0.0391 rooms. This comparison includes equilibrium and risk-set changes.

The gain has limits: mean rooms rise to 6.349, and the parent ownership gap widens to 0.1757 against 0.1677. The first-birth response still falls short by **0.2045 rooms**. This supports joint housing-and-tenure refinement while retaining all twelve moments; it does not close the calibration problem.

All estimated parameters and restrictions

Parameter	Bounded estimate	Search interval	Near bound?
Annual patience β	0.995739	[0.94, 0.9995]	No
First-birth dispersion κ_1	2.232730	[0.02, 50]	No
Later-birth dispersion κ_C	1.736471	[0.02, 50]	No
Owner premium χ	1.043472	[0.1, 5]	No
Supply intercept H_0	14.562959	[0.2, 80]	No
Bequest strength θ_0	0.554442	[0, 8]	No
Bequest shifter θ_1	0.106357	[0.02, 16]	Raw scale
Per-child floor $\bar{h}_C$	0.237187	[0.1, 1.8]	No
First-birth fixed cost	4.544053	[0, 8]	No
First-child term $\bar{h}_J$	0.500000	[0, 0.5]	Upper ceiling
Dated child-value change	-0.326881	[-1.5, 0.2]	No

In the three-child-floor-preserving tests, **all nine coordinates other than the two housing floors retain the values above**. The first-child term is externally fixed at the diagnostic value; the slope is set to preserve the three-child floor. Neither is estimated in this exercise. The slope at jump 0.9 is near its original lower bound; the first-child values are outside the original [0, 0.5] interval.

Fixed first-child term	Per-child slope	Normalized old intercept	Derived 2023 intercept
0.600000	0.203854	0.297403	-0.029477
0.750000	0.153854	0.301649	-0.025232
0.900000	0.103854	0.305974	-0.020907

The near-bound flag follows the stored rule: within 2% of the range in raw parameter units. The bequest shifter can therefore be flagged even when it is well inside its logarithmic search coordinate. The dated housing-supply elasticity is fixed at 0.63 and tenure-choice dispersion at 0.005. The inherited old-equilibrium initialization still uses elasticity 1.75. The old fertility intercept is normalized to completed fertility 2.1; the dated preference change is an estimated coordinate. None of these restrictions is newly estimated here.

At the bounded winner, the normalized old child-value intercept is 0.294616 and the derived 2023 intercept is -0.032265. The young-ownership validation row remains outside the objective: the retained whole-node convention gives 31.094% versus the ACS 34.117% target. The coarse age-cell measurement is recorded separately in the diagnostic packet.

What the numerical checks establish

The evening work completed **50 full historical evaluations**: 39 in the bounded search, five in the first diagnostic and six in the observed-room-gap follow-up. The same source, target fingerprint, population/accounting contract, market tolerance and feasibility checks pass. Each history has a 2023 checkpoint and the unchanged seventeen-graph packet. There were also two fixed-price Bellman evaluations for the value warning described below. Allocated cluster time was **8.05 CPU-hours**.

The collector verified all **1,050 originally recorded artifact hashes** on Torch. All **2,742 collected local files** match the final manifest; **52 large checkpoints** remain on Torch. The bounded final repeats, fixed-option smokes and final diagnostic repeats reproduce all twelve fit rows, all physical parameters, 253 numeric historical entries and all seventeen standard PNGs exactly. The five launcher / planning guard checks pass.

The bounded winner and all seven extended diagnostic points have zero occupied value decreases. Maximum reported-budget excess share across the histories is 1.19×10^{-8} . Inspection of both selected points and the largest extension preserves the familiar equilibrium shapes: ownership still approaches one at old ages, and tenure policies remain uneven near some wealth thresholds. A lower objective does not settle these validation issues.

The coordinate panel around the morning starting point gives a weighted Jacobian with condition number **1,335**. It has eleven nonzero singular values, but only ten exceed the relative 10^{-3} screening threshold. Its weakest direction is dominated by the bequest shifter. This is a warning about local conditioning under the chosen step and parameter scaling; it is not proof of global underidentification or an identification certificate for the new winner.

The main disagreement between upward and downward derivatives comes from the old-age wealth-to-income p90 / p50 target. Nine saved-state reconstructions reproduce that statistic exactly. The code uses an inverse-CDF quantile on discrete support: a small discount-factor reduction moves p90 from 27.895 to 27.587 while p50 stays near 8.443. This explains an objective jump without implying a coding error. The housing-floor derivatives are much more consistent. No moment was dropped, smoothed or reweighted. The full Jacobian was not inverted; the final four-point diagnostic used only the stable floor derivatives to choose explicit test points.

One rejected tenure-preference case has a value decrease over an adjacent wealth step, affecting 0.229 per million of pre-choice mass. A fresh local solve reproduces its policies and distributions exactly. Exhaustive saving removes the decrease; at the same prices and distribution, births change by 0.000234% and ownership by -0.000022 percentage points. This does not bound the error of a re-solved historical calibration or of the matched birth-rooms statistic. The verified quantity differences are too small to account for the large ownership movement at that saved state.

Policy context and next discussion

These policy numbers were verified at the **morning candidate, loss 23.153**. They have not been re-computed at tonight's new points. Entries are percent changes in births per household against that candidate's own no-policy path.

Policy	Impact, 2023	2063
Supply +20%	+1.343%	+2.141%
Dependent-child LTV 95%	+0.018%	+0.075%
Supply and credit	+1.366%	+2.244%
Property tax 1% to 2%, no rebate	-1.145%	-1.699%

The separate rebated 2%-versus-rebated 1% impact decomposition is -1.377% direct tax, +0.302% capitalization and +1.574% rebate, totaling +0.500%. Its fiscal baseline differs from the unrebated tax row. These remain conditional mechanism comparisons, with an inherited household-entry closure; they are neither welfare results nor validated population forecasts.

The next discussion should settle three things: whether to enlarge the first-child parameter's search interval on the evidence above; which candidate is ready for the September 14 presentation; and which population-closure assumptions can support the policy interpretation. Keep the full twelve-row objective in the next search. The current evidence does not establish that the first-birth response is unreachable.

The empirical reference-choice issue remains a documentation item. The completed paired regression reproduces 0.720246 rooms; changing one cohort reference gives 0.730459 with identical observations and fitted values. This demonstrates sensitivity of the aggregation convention, but the tested change is small and upward. It supplies no basis for lowering the calibration target. The pooled empirical cohort mix and matched model branch should also be described distinctly.

Evidence and reproduction

The [experiment README](#) records the design, deadlines, jobs, immutable source and target hashes, and graph replay command. The [complete bounded-case ledger](#), all twelve-row fits and all parameter tables include the rejected points.

Earlier comparison points retain their complete tables: [production fit](#), [production parameters](#), [morning fit](#), and [morning parameters](#). The [earlier policy review](#) contains the full policy evidence. The [empirical measurement review](#) records the regression replay and its limits.

The [conditional profile ledger](#), [complete profile fits](#), [profile parameters](#) and [stop receipt](#) record the extended points and why further identical repeats were omitted.

The [final four-point ledger](#), [full fits](#) and [parameter ledger](#) record the observed-room-gap follow-up. Its preparation and corrected collector are saved in the experiment folder.

The standard seventeen diagnostic plots accompany every completed historical evaluation. The saved-state graph replay reproduced all seventeen original PNG hashes exactly with no model solve. The [experiment README](#) gives the single regeneration command, all plans, budgets and source restrictions. Production remains the September 4 `task_010`; author-controlled manuscript files were not edited.